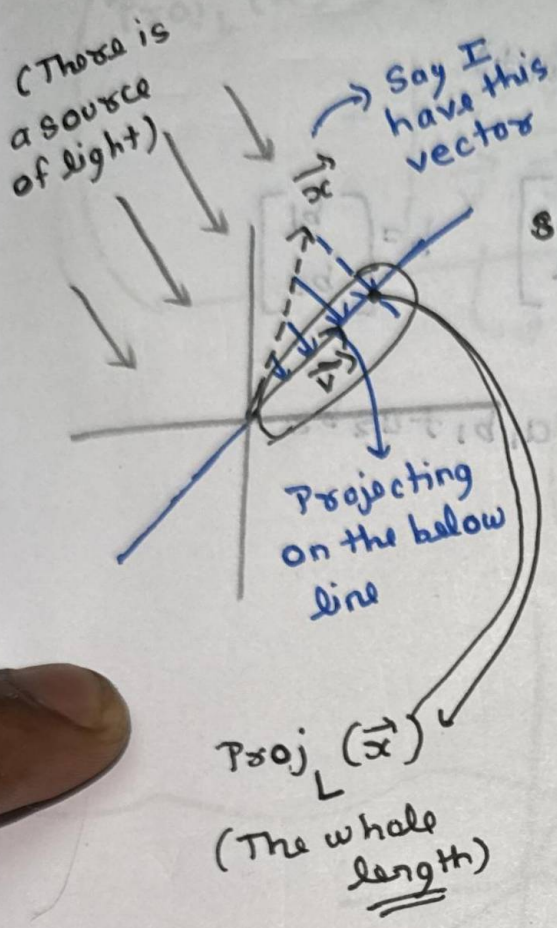
17. Introduction to Projection



Say I can create a line (L) with the vector ($\vec{v}$)

$$L = \{c\vec{v} \mid c \in \mathbb{R}\}$$

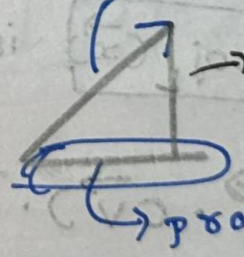
if you multiply the vector with a scalar (Extend it to a straight line)

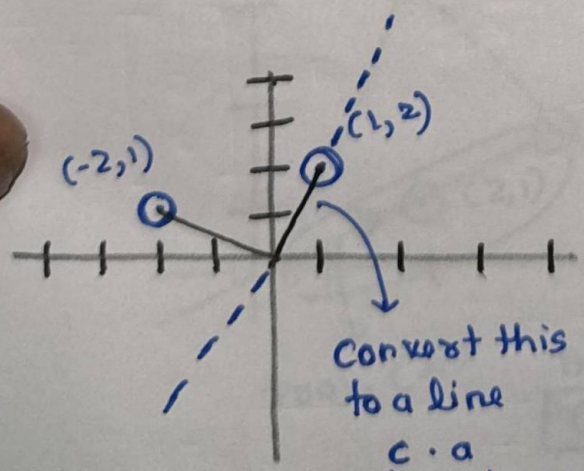
$Proj_L(\vec{x}) \rightarrow$ Projection of $\vec{x}$ on the line (L)

$\boxed{Proj_L(\vec{x}) \rightarrow \text{Some vetor in Line}}$

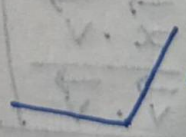
This $\vec{x} - Proj_L(\vec{x})$ is $\perp$r to L

So,





$$a = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad b = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$



(At this state both of them are $\perp$r to each other)

Now,

$$a \cdot b = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

(Doing the DOT Product)

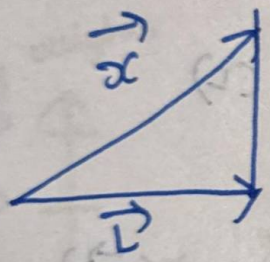
$\hookrightarrow a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

$a \cdot b = (1)(-2)$

$a \cdot b = a_1 b_1 + a_2 b_2$

$\boxed{a \cdot b = 0}$

$\hookrightarrow$ DOT PRODUCT of $\perp$r vectors is $\boxed{0}$



$\vec{x} - \text{Proj}_L(\vec{x})$

This is basically $c\vec{v} \rightarrow$ vector. (Scalar) becomes a line

Now

$\vec{x} - \boxed{\text{Proj}_L(\vec{x})}$ is $\perp$r to L

$$(\vec{x} \cdot c\vec{v}) \cdot \vec{v}$$

$$\vec{x} \cdot \vec{v} - c\vec{v} \cdot \vec{v} = 0$$

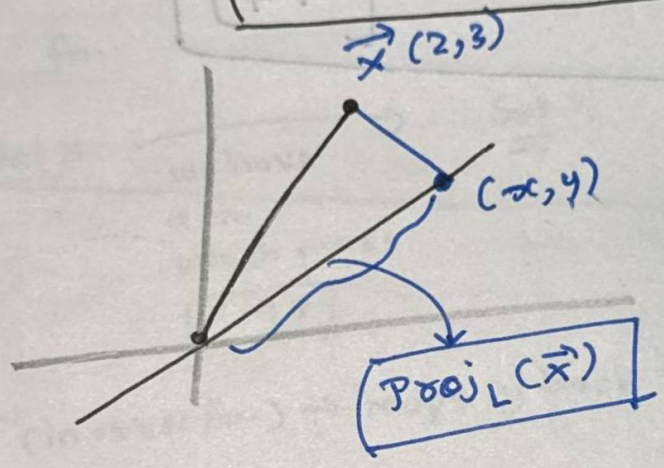
$$\vec{x} \cdot \vec{v} = c\vec{v} \cdot \vec{v}$$

$$\boxed{c = \frac{\vec{x} \cdot \vec{v}}{\vec{v} \cdot \vec{v}}}$$

$$\boxed{\begin{aligned} \text{Proj}_L(\vec{x}) &= c\vec{v} \\ &= \left| \frac{\vec{x}\cdot\vec{v}}{\vec{v}\cdot\vec{v}} \right| \cdot \vec{v} \end{aligned}}$$

In books we see it to be

$$\boxed{\text{Proj}_b(a) = \left(\frac{a\cdot b}{b\cdot b}\right)\cdot b}$$

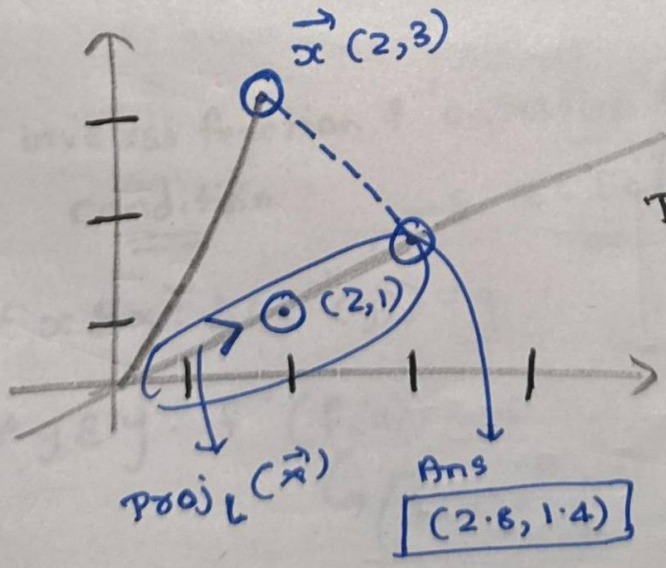


Example

$$L = \left\{ c\begin{bmatrix}2\\1\end{bmatrix} \quad c \in \mathbb{R} \right\}$$

$c\vec{v}$

$$\boxed{\vec{x} = \begin{bmatrix}2\\3\end{bmatrix}}$$



$$\text{Proj}_L(\vec{x}) = \left(\frac{\begin{bmatrix}2\\3\end{bmatrix}\cdot\begin{bmatrix}2\\1\end{bmatrix}}{\begin{bmatrix}2\\1\end{bmatrix}\cdot\begin{bmatrix}2\\1\end{bmatrix}}\right)\cdot\begin{bmatrix}2\\1\end{bmatrix}$$

$$\Rightarrow \frac{4+3}{4+1} \Rightarrow \frac{7}{5} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\cancel{\Rightarrow 14+8} \Rightarrow \begin{bmatrix} 14/5 \\ 7/5 \end{bmatrix}$$

$$\cancel{\rightarrow \boxed{19}}$$

this is not a vector (Post vector calculation it is a scalar)

$$\boxed{\text{Proj}_L(\vec{x}) \Rightarrow \begin{bmatrix} 2.8 \\ 1.4 \end{bmatrix}}$$